Jacobian Descent For Multi-Objective Optimization

Valérian Rey — Simplex Lab
valerian.rey@gmail.com
Joint work with Pierre Quinton

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GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

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Theory

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

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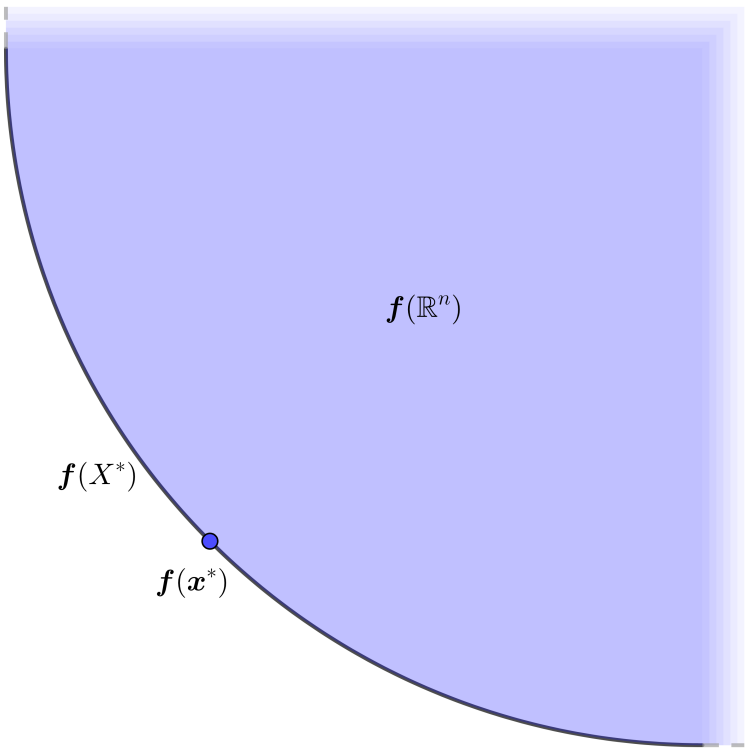

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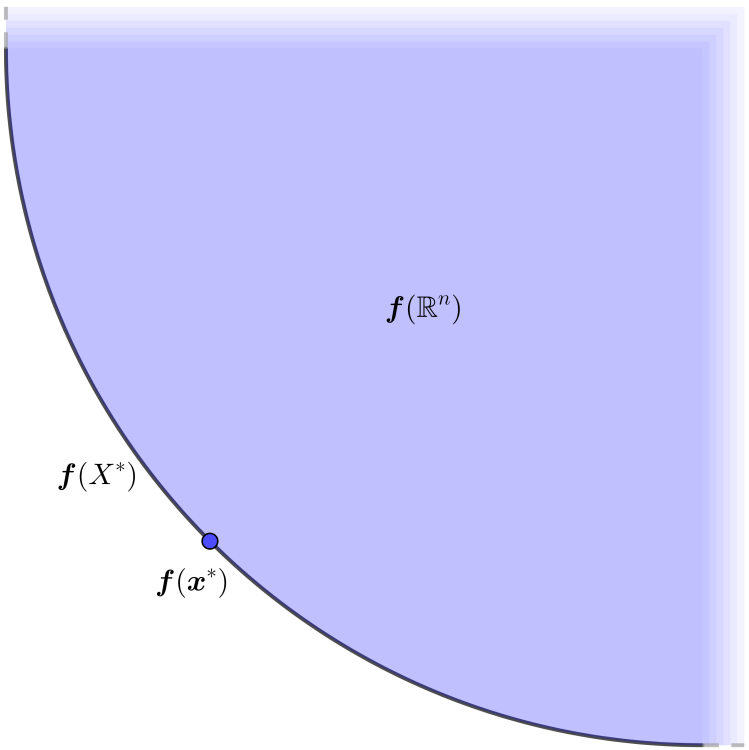

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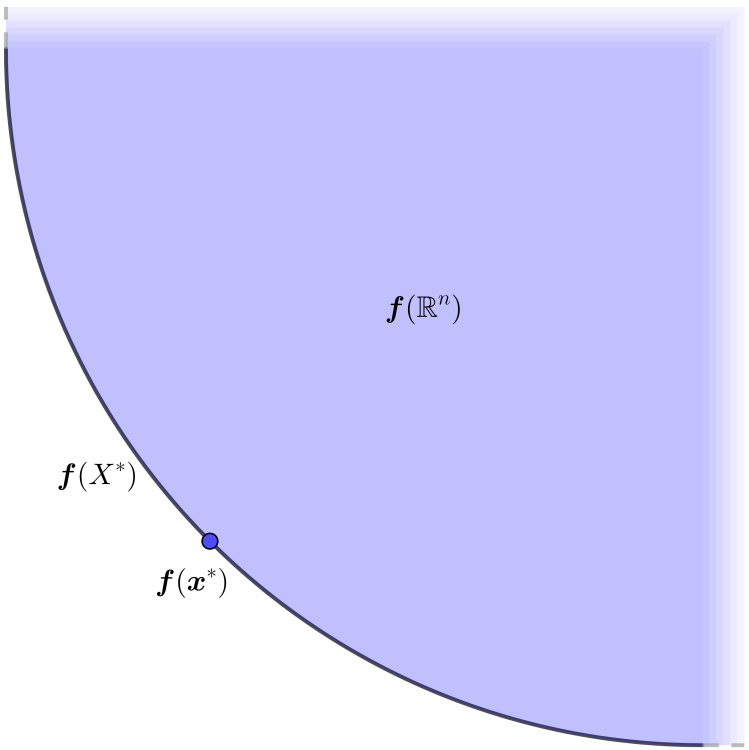
Pareto Front: $\boldsymbol{f}(X^*)$



First-order Taylor approximation

For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

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$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

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$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

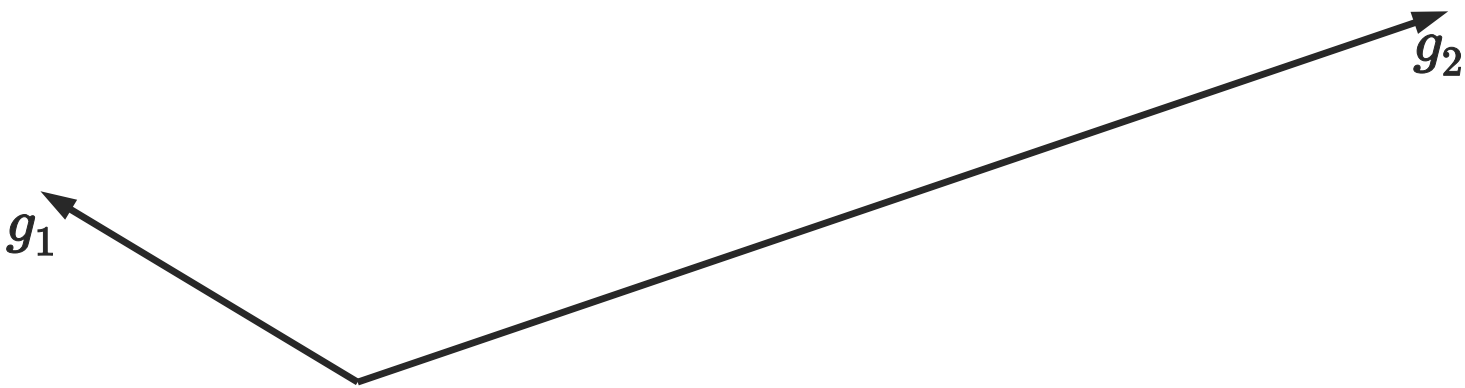
Input: $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$

for $t \leftarrow 1$ **to** T **do**

$\quad \boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$

Output: $\boldsymbol{x}$

Unconflicting projection of gradients (UPGrad)



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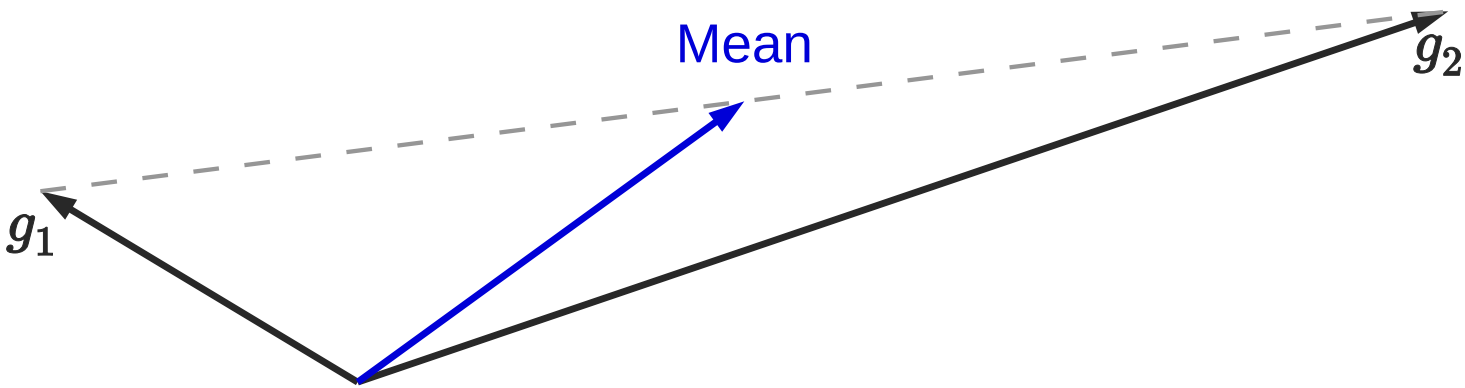

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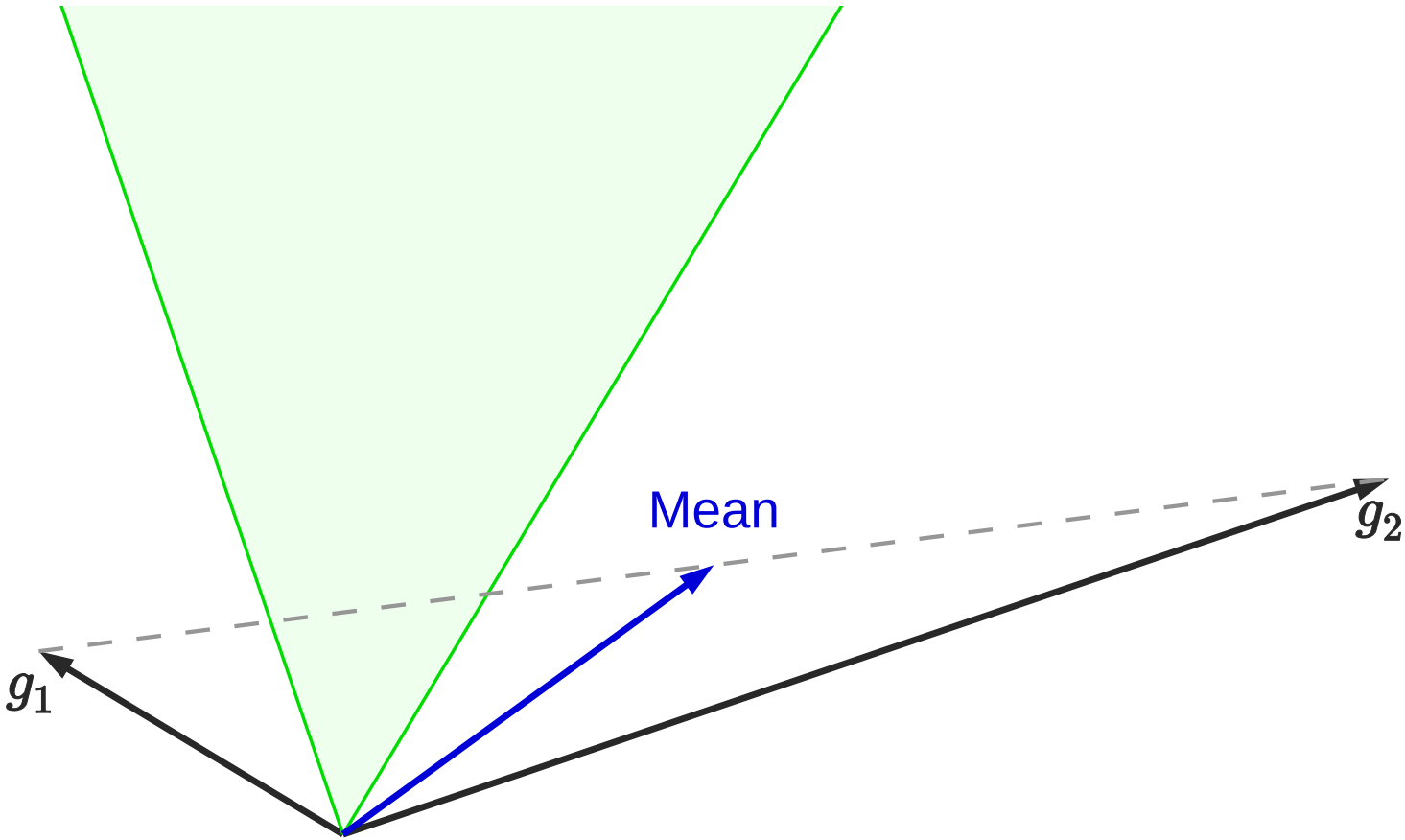

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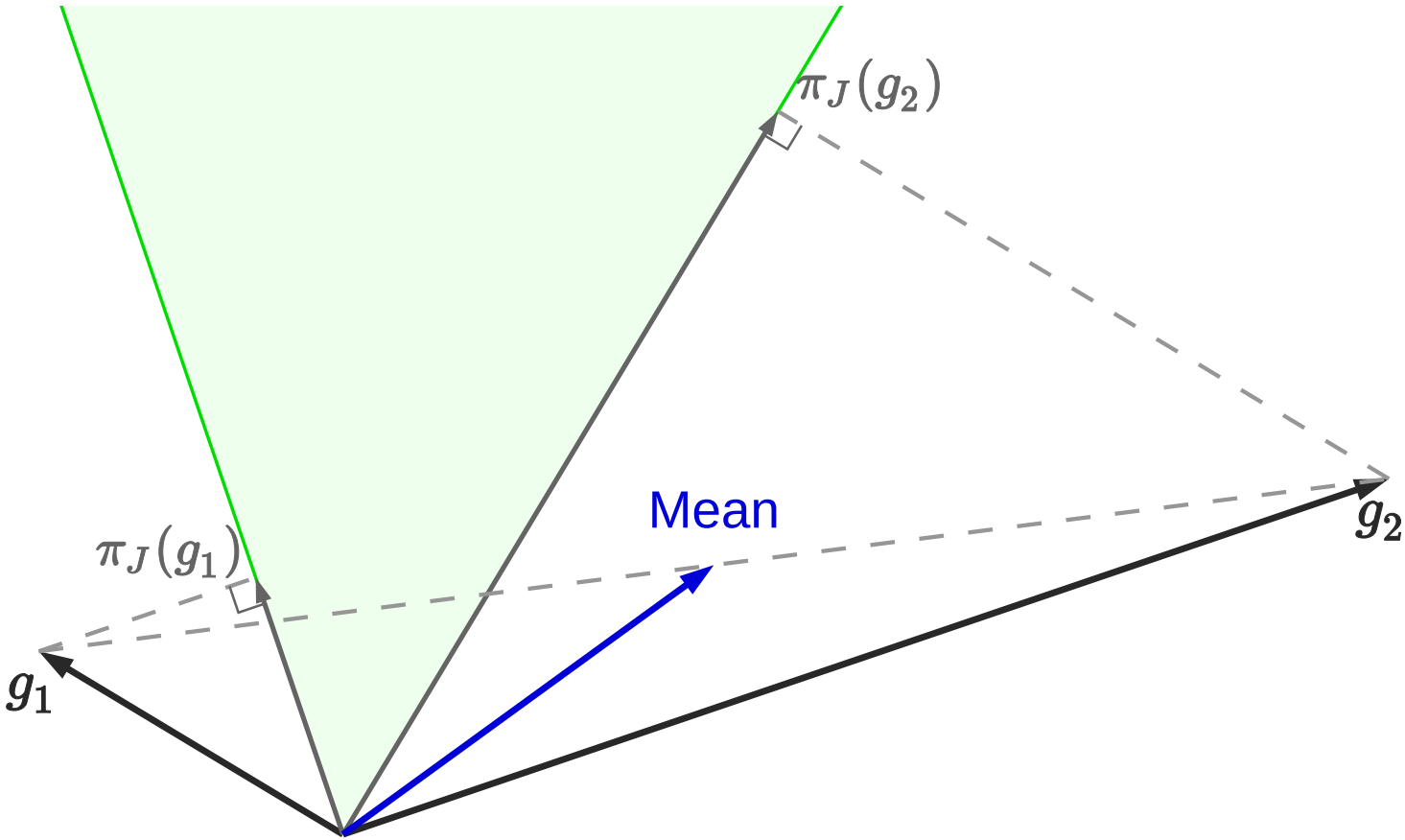

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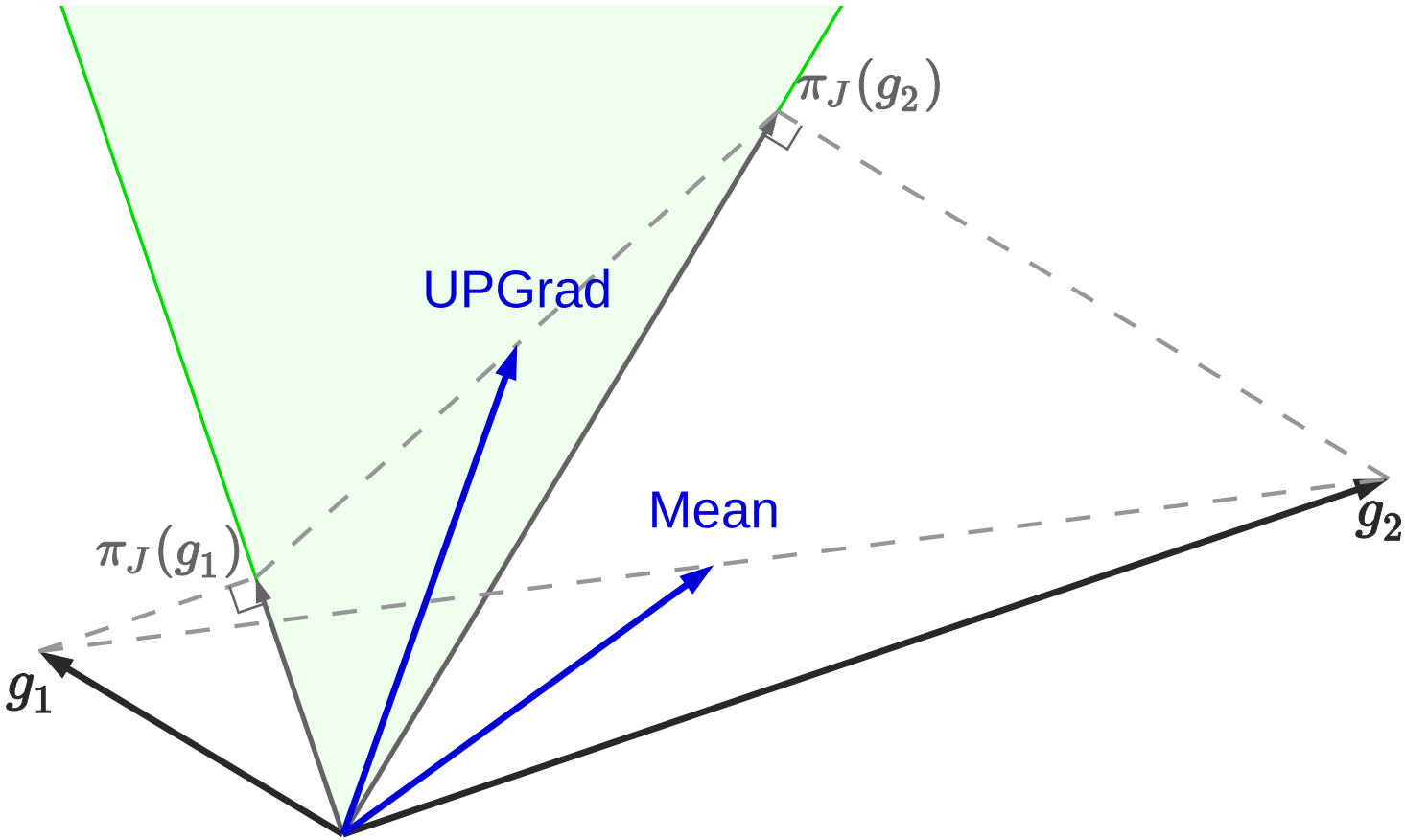


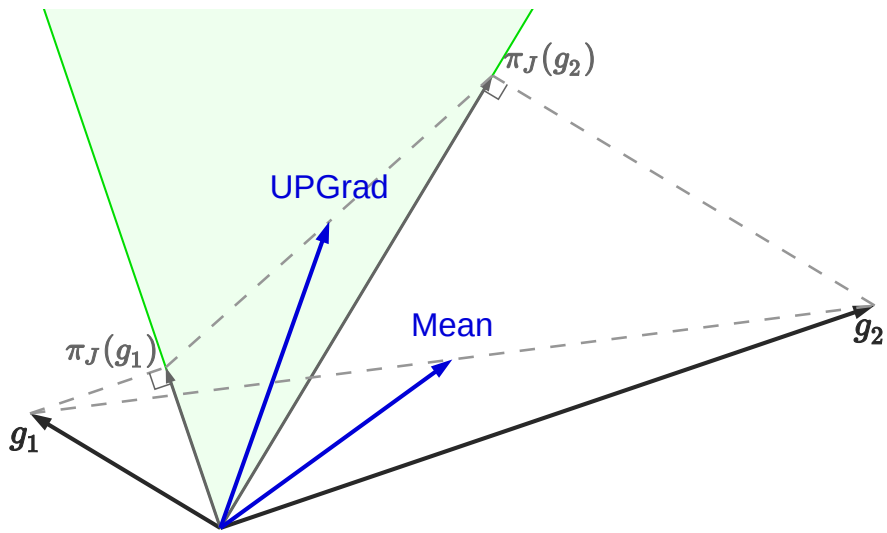
Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{0} \preceq J\boldsymbol{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m \times n}$, $\mathcal{A}(J) \in C_J$.

For any $\boldsymbol{x} \in \mathbb{R}^n$, the projection of $\boldsymbol{x}$ onto C_J is

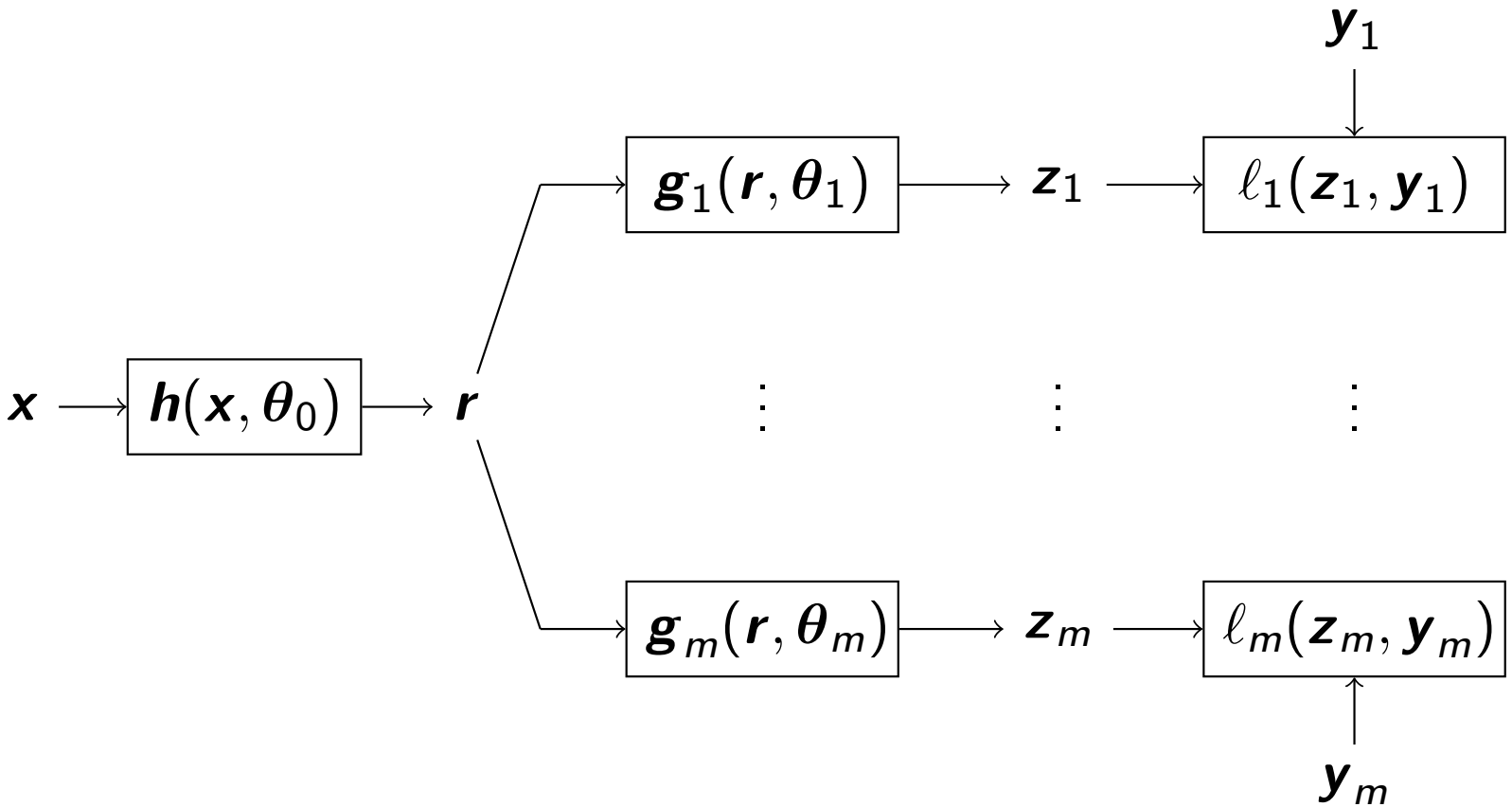
$$\pi_J(\boldsymbol{x}) = \underset{\boldsymbol{y} \in C_J}{\arg\min} \|\boldsymbol{x} - \boldsymbol{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J\left(J^\top \boldsymbol{e}_i\right)$$



Applications and experimentation

Multi-task learning



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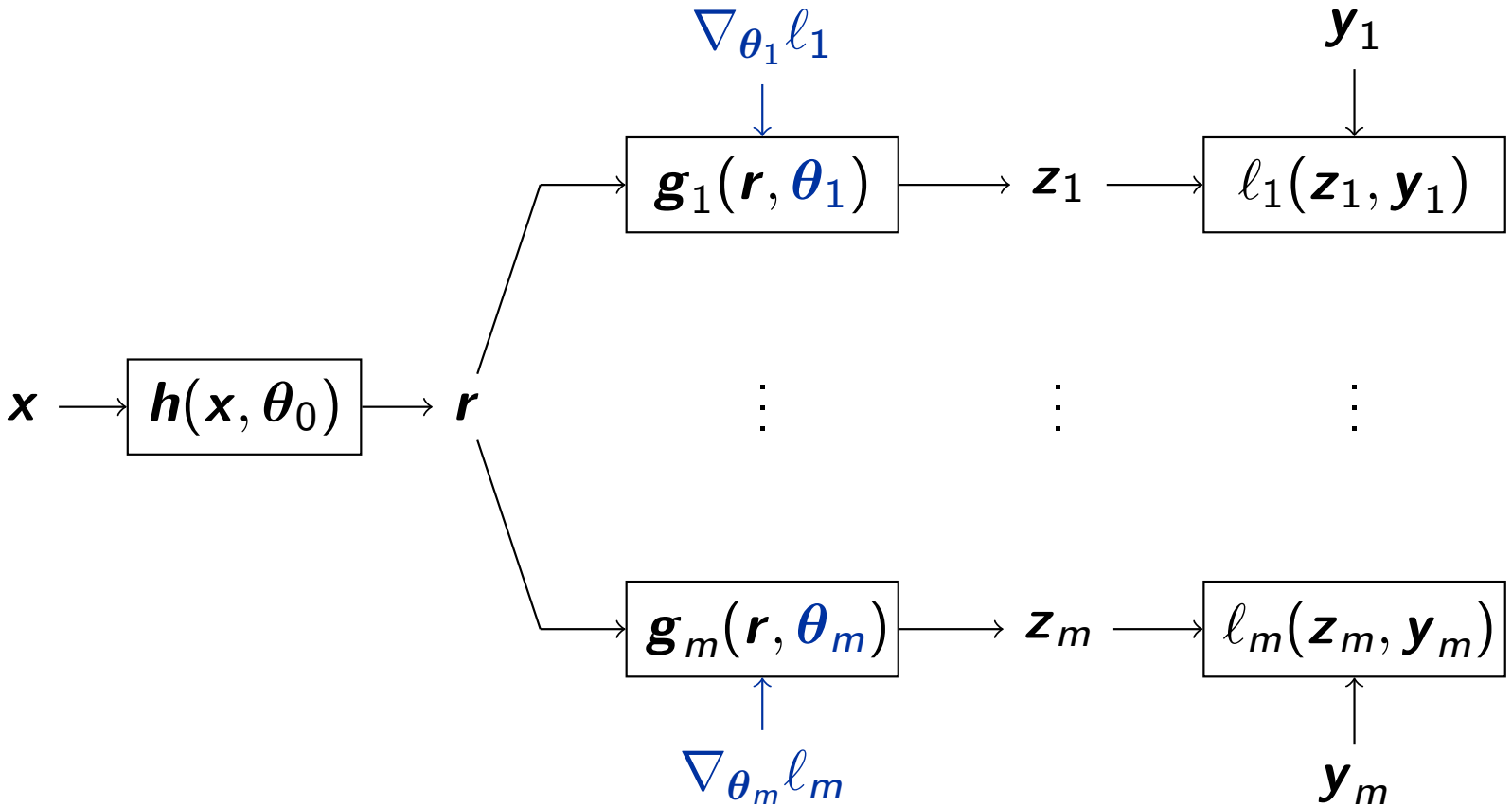

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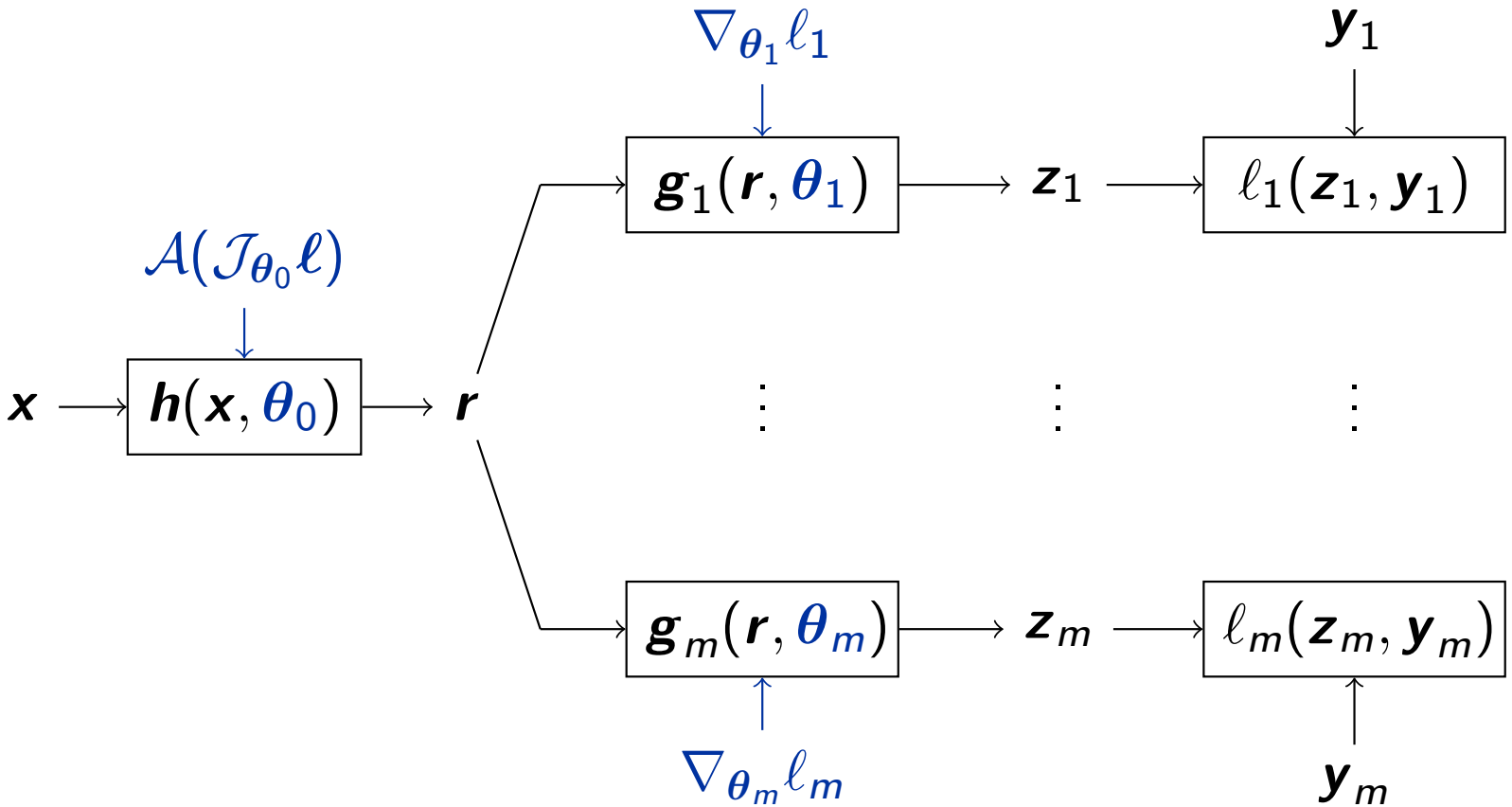

Experimental results: per-sample JD

Each element of the batch has its own loss.

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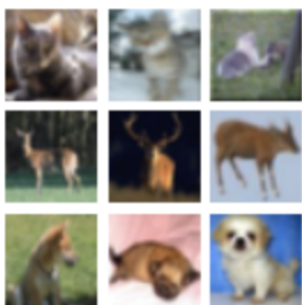
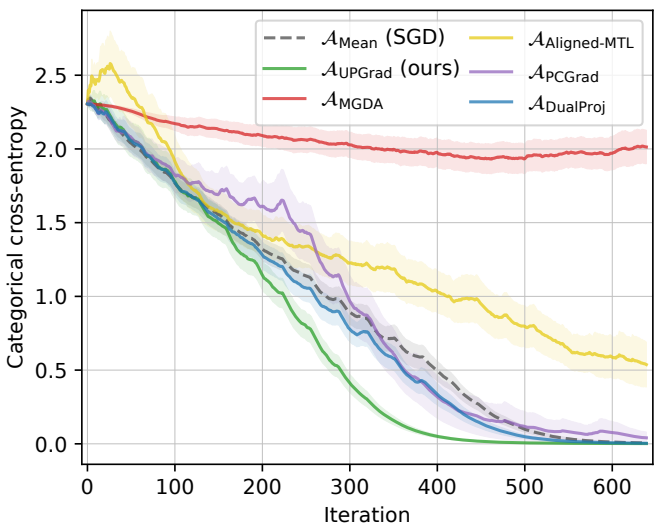
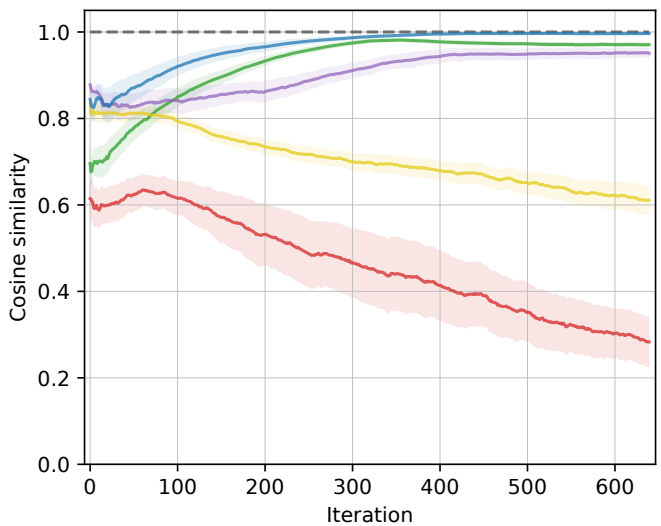
CIFAR-10



Training Loss



Update Similarity



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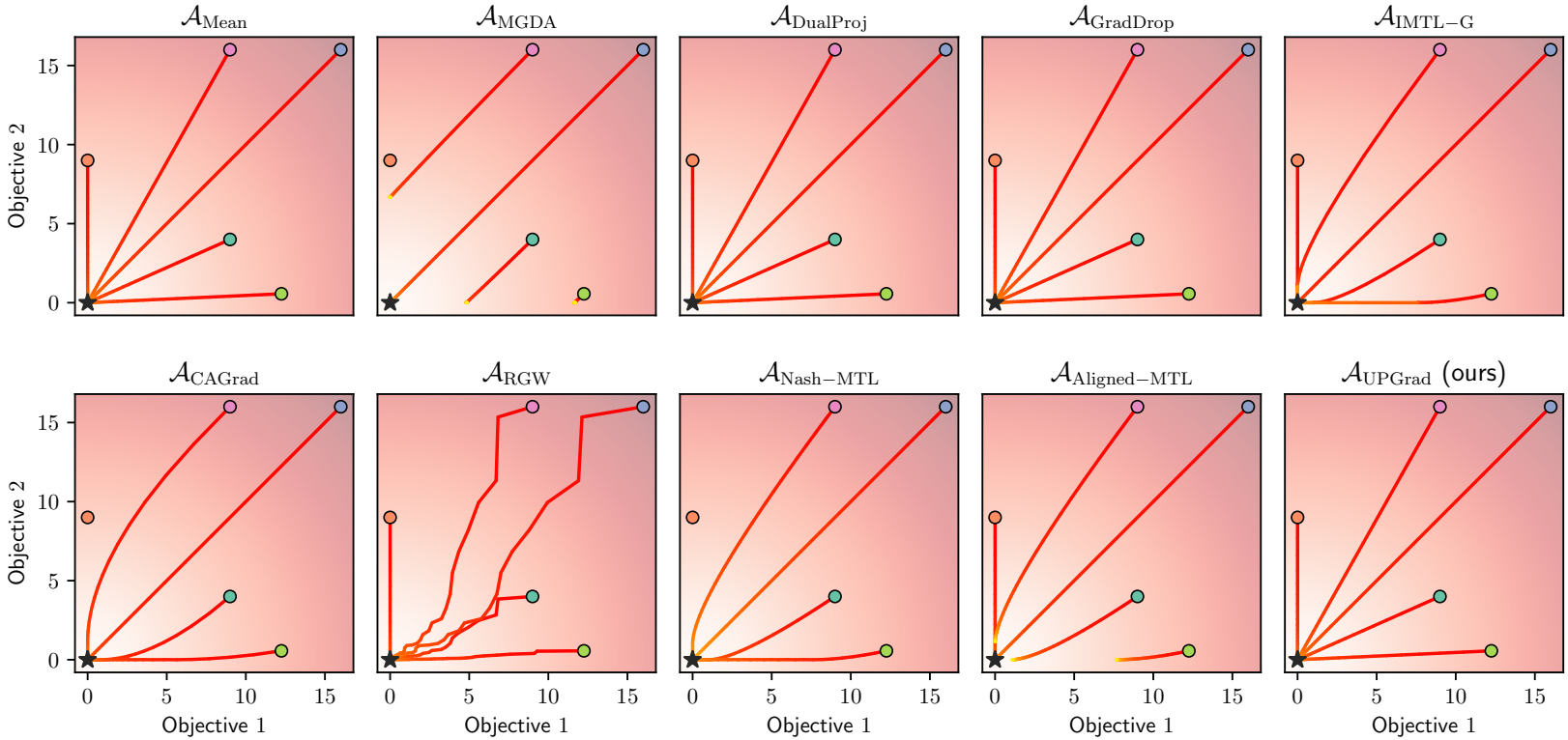
- Find new applications.

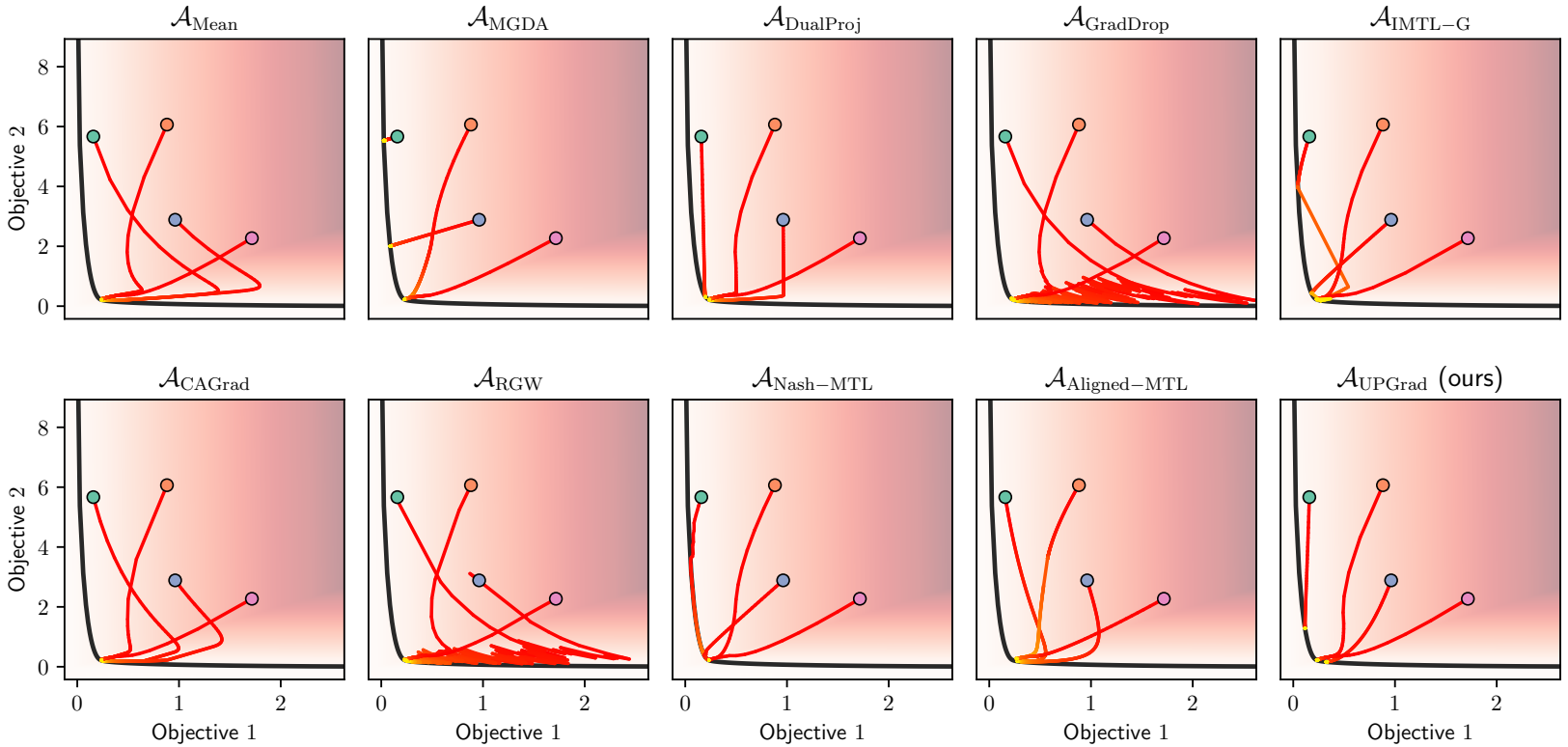
Bonus: Element-wise quadratic function trajectories

Minimize $\boldsymbol{f}$: $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.



Bonus: Convex quadratic form trajectories

Some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with highly conflicting gradients.



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Applicable to most other aggregators from the literature.